# Successive Interference Cancellation in a Narrow-Beam LEO Uplink...

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***Abstract*—We investigate the order statistics of the signal-to-interference ratios (SIRs) at a typical low Earth orbit satellite base station (SBS) with a narrow Gaussian antenna serving an urban area, using a Gaussian mixture shadowing model. We analyze the coverage probability at the typical SBS receiving the three strongest signals and implement successive interference cancellation (SIC). The signal-to-total-interference (STIR) process is characterized by a Poisson-Dirichlet distribution PD$(0,\cdot)$, which allows us to derive the density of the factorial moment measure. In the results, we demonstrate that by SIC, we can, for example, make the network twice as dense while preserving the performance of the strongest user equipment (UE) and significantly improving the performance of the second- and third-strongest transmitters.**

## I. Introduction and motivation

Successive interference cancellation (SIC) has the potential to improve the performance of multiple access communication systems. The order statistics of the signal-to-interference ratio (SIR) under interference cancellation (and signal combination) have been studied for terrestrial networks [**?**] using stochastic geometry, they have yet to be explored in low Earth orbit (LEO) networks, where the topology is remarkably different. A SIC scheme was studied in [**?**] in the stochastic geometry context in satellite networks.

We study the SIR of user equipments (UEs) at a typical LEO satellite base station (SBS) by utilizing the narrow-beam LEO uplink system model from [**?**] in an urban environment. We utilize the Gaussian Mixture shadowing model, similar to [**?**] and [**?**], using the parameters presented in [**?**]. The density of the factorial moment measure of the signal-to-total-interference (STIR) process follows a Poisson-Dirichlet distribution PD$(0,\cdot)$ (contrary to PD$(\cdot,0)$ in [**?**]), of which the factorial moment measure is well-known. We derive the joint PDF of the STIR and SIR and study the performance metrics of the three strongest UE signals with and without SIC. In this paper, we demonstrate that the link quality is more stable with interference cancellation because we can increase the network (and constellation) density while mitigating interference with the SIC. Furthermore, the coverage probability of UEs with weaker signals drastically improves, so a single SBS can serve multiple UEs.

Furthermore, in the paper, a highly simplified moment-order-two approximation for log-normal shadowing is proposed and shown to be feasible, with a worst-case error in the point-wise SIR distribution approximation $< 0.2$: a *mixture* exponential distribution with a delta at 0 (*two* degrees of freedom instead of *one*). The exponential fading distribution typically yields the simplest and most insightful results in stochastic geometry analysis, which can be leveraged by this approximation.

The work was supported by the Research Council of Finland Grant 339446.

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TABLE I: Principal symbols the values and units in the numerical results. We denote (approximate) proportionality "$\propto$" or equality "$=$" to a variable. For a dimensional number, we denote the units "SI" and "[non-SI]."

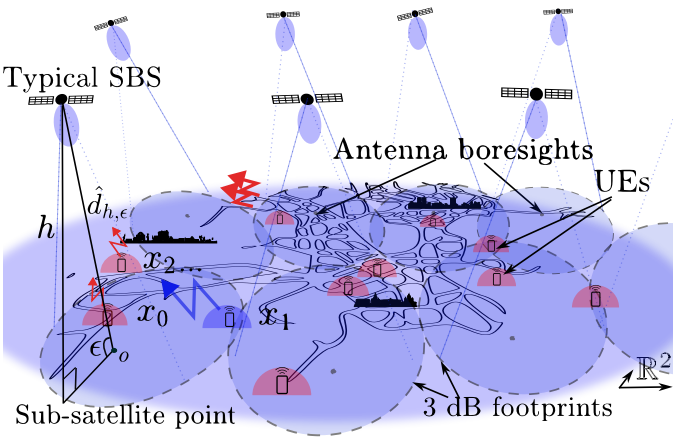
| Symbol | Explanation | Values and units |
|---|---|---|
| $h$ | Receiving SBS altitudes. | 1000 km |
| $\alpha$ | Power path loss exponent. | 4 |
| $\lambda$ | The density of $\Phi$ and $\Theta$, *i.e.*, the mean number of UEs inside an unit area. | $\{0.83, 13.3\} \times 10^{-4}$/km$^2$ |
| $\epsilon$ | The typical SBS elevation angle w.r.t $o$. | $(\pi/2, \pi/6)$ rad $= (90, 30)[^\circ]$ |
| $p_{\text{LoS}}$ | LoS probability; $p_{\text{LoS}} \propto \sin(\epsilon)$. | $(0.992, 0.493)$ |
| $\mu_{\text{LoS}}$ | Mean of the LoS component of the Gaussian mixture shadow fading model. | 0 [dB] |
| $\mu_{\text{NLoS}}$ | Mean of the NLoS component. | $-26$ [dB] |
| $\sigma^2_{\text{LoS}}$ | Variance of the LoS component. | $4^2$ [dB] |
| $\sigma^2_{\text{NLoS}}$ | Variance of the NLoS component. | $6^2$ [dB] |
| $\varphi_{\text{RX}}$ | Half-width of the $-3$ dB antenna gain. | $0.028 = 1.6[^\circ]$ |
| $\kappa$ | Average number of UEs inside a $-3$ dB footprint; $\kappa = \pi\lambda\left(\varphi_{\text{RX}} h/\sin^2(\epsilon)\right)^2$. | $(1.63, 6.51)$ |
| $\upsilon$ | Fraction of effective UEs; $\upsilon \propto \sin(\epsilon)$. | $(0.85, 0.426)$ |
| $\kappa\upsilon$ | Average number of effective UEs inside a $-3$ dB footprint. | $\{2, 4\} \times \log(2)$ |
| $\tilde{\kappa}$ | $\kappa/\log(2)$. | |
| $\theta$ | SIR threshold of a successful transmission. | $(0.2, 10) = (-7, 10)$ [dB] |
| $\tau$ | SIR threshold of a successful interference cancellation. | $0.2 = -7$ [dB] |

## II. System model

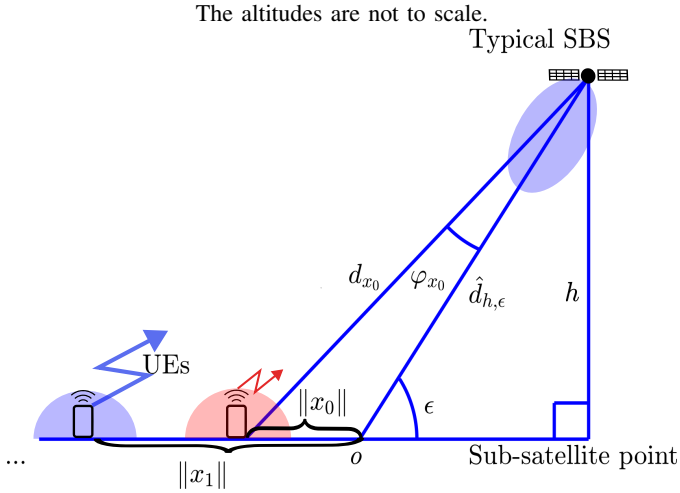
A narrowband LEO uplink is considered with narrow-beamed LEO receivers, and statistics over short use periods are studied. Due to the narrow beam, the Doppler spread can be ignored. The UEs follow a homogeneous PPP $\Phi \subset \mathbb{R}^2$ of density $\lambda$. No fast-fading is modeled, and the SBSs are associated with the served UEs according to their signal powers at the SBS after log-normal shadowing. The SBSs form a homogeneous point process (p.p.), possibly with a different density than the UEs. Because of the translation invariance of the PPP, all locations are statistically equivalent, and we define the origin $o$ to represent the typical SBS footprint focus point.

The path loss, representing the antenna gain at the typical SBS, is given over the planar distance $r \in [0, \infty)$ as a Gaussian function

$$G(r) = 2^{-(D_{h,\epsilon} r)^2/\varphi^2_{\text{RX}}}. \tag{1}$$




(a) Interpretation of the *planar* system model with the satellites in adjacent orbits serving an urban area and a realization of the UEs. The altitudes are not to scale.




(b) The typical SBS as seen from the side. The transmitters are projected into line $(0, \infty)$ according to their norm.

Fig. 1: The simplified narrow-beam LEO uplink system model. The satellite antenna boresight is oriented towards $o$, the focus point of the elliptical footprint. The omnidirectionally transmitting UEs $\{x_i\}$ are located according to the homogeneous PPP on the plane. The transmitter with the strongest signal is the first-served UE.

The angle $\varphi_{\text{RX}}$ denotes the $-3$ dB antenna gain width. Furthermore, the scaling constant $D_{h,\epsilon} \triangleq \sin^2(\epsilon)/h$ is a first-order coefficient of the Taylor expansion of the angle $\varphi_r$ w.r.t. the boresight of the typical antenna pattern. (Thorough details in [**?**].)

### A. Shadowing

#### 1) Gaussian mixture shadowing model

Consider a two-tier $\{\text{LoS}, \text{NLoS}\}$ (line-of-sight and non-line-of-sight) Gaussian mixture shadow fading model with the parameters $\mu_{\text{LoS}} = 0$ dB, $\sigma_{\text{LoS}} = 4$ dB, $\mu_{\text{NLoS}} = -26$ dB, and $\sigma_{\text{NLoS}} = 6$ dB, which correspond to an urban environment [**?**]. Assuming i.i.d. power shadow fading for all UEs, *the typical shadowed transmit power* $H_{\mathcal{MLN}}$ follows a log-normal mixture distribution;

$$H_{\mathcal{MLN}} \sim p_{\text{LoS}} \mathcal{LN}(\rho\mu_{\text{LoS}}, (\rho\sigma_{\text{LoS}})^2) + p_{\text{NLoS}} \mathcal{LN}(\rho\mu_{\text{NLoS}}, (\rho\sigma_{\text{NLoS}})^2), \quad (2)$$

where $p_{\text{LoS}} = 1 - p_{\text{NLoS}}$ is the LoS probability. Considering a natural base for the log-normal distribution, the constant $\rho \triangleq \log(10)/10$ normalizes the parameters $\mu_{\text{LoS}}, \sigma_{\text{LoS}}, \mu_{\text{NLoS}}$, and $\sigma_{\text{NLoS}}$, ensuring that the conditioned r.v.'s $10\log_{10}(H_{\mathcal{MLN}}|\text{LoS})$ and $10\log_{10}(H_{\mathcal{MLN}}|\text{NLoS})$ evaluate to r.v.'s following the normal distributions $\mathcal{N}(\mu_{\text{LoS}}, \sigma^2_{\text{LoS}})$ and $\mathcal{N}(\mu_{\text{NLoS}}, \sigma^2_{\text{NLoS}})$, respectively.

#### 2) Mixture exponential shadowing distribution

As a compromise between analytical tractability and realism, we introduce an exponential distribution with an atomic probability measure at 0, described by the complementary CDF (CCDF)

$$F_{H_{\text{Exp}}}(t) = \upsilon e^{-t}, t > 0. \quad (3)$$

The parameter $0 \leq 1 - \upsilon < 1$ denotes the probability that the shadowed signal is entirely attenuated and takes the value of zero, otherwise, the power follows the exponential distribution.

In the following, the scaling term $\Upsilon$ ensures that the means of the log-normal mixture distribution and the mixture exponential distribution match: $\mathbb{E}(\Upsilon \times H_{\mathcal{MLN}}) = \mathbb{E}(H_{\text{Exp}}) = \upsilon$. By equating the first two moments M1 and M2 of $H_{\text{Exp}}$ (l.h.s.) and $\Upsilon \times H_{\mathcal{MLN}}$ (r.h.s.);

$$\begin{cases} \text{M1: } \upsilon = \Upsilon \left( p_{\text{LoS}} e^{\mu_{\text{LoS}} + \sigma^2_{\text{LoS}}/2} + p_{\text{NLoS}} e^{\mu_{\text{NLoS}} + \sigma^2_{\text{NLoS}}/2} \right) \\ \text{M2: } 2\upsilon = \Upsilon^2 \left( p_{\text{LoS}} e^{2(\mu_{\text{LoS}} + \sigma^2_{\text{LoS}})} + p_{\text{NLoS}} e^{2(\mu_{\text{NLoS}} + \sigma^2_{\text{NLoS}})} \right), \end{cases} \quad (4)$$

we can solve for the parameter $\upsilon$ for the shadowing fading distribution approximation $F_{H_{\text{exp}}}(\cdot)$:

$$\upsilon = \frac{2\left( p_{\text{LoS}} e^{\mu_{\text{LoS}} + \sigma^2_{\text{LoS}}/2} + p_{\text{NLoS}} e^{\mu_{\text{NLoS}} + \sigma^2_{\text{NLoS}}/2} \right)^2}{p_{\text{LoS}} e^{2(\mu_{\text{LoS}} + \sigma^2_{\text{LoS}})} + p_{\text{NLoS}} e^{2(\mu_{\text{NLoS}} + \sigma^2_{\text{NLoS}})}}. \quad (5)$$

The parameter $\upsilon = \upsilon(\epsilon)$ varies with the elevation angle through the LoS probability, which influences the shadow fading characteristics. The parameter $\Upsilon$ holds no significance in an interference-limited scenario, as the equal scaling of all UE powers neutralizes its effect.

**Remark.** *The variable $0 < \upsilon \leq 1$ is not generally solvable for all log-normal distribution parameters (by matching the first two moments). Broadly speaking, the variance of the shadowing has to be large enough. However, the variable $0 < \upsilon \leq 1$ is solvable for almost every shadowing scenario in [**?**], particularly for the urban scenario.*

### B. The spherical Earth model in the Monte Carlo simulations

The planar model approximates the spherical system model, where the UEs are located on the spherical Earth surface of radius $R_\oplus = 6378$ km according to a homogeneous PPP $\Theta$ of density $\lambda$ represented in spherical coordinates. This p.p. can be constructed from $\Phi \subset \mathbb{R}^2$ by an area preserving mapping. Namely, for $x = (x_1, x_2) \in [-\pi, \pi] \times [-1, 1] \cap \Phi/R_\oplus$,

$$(x_1, x_2) \mapsto (R_\oplus, x_1, \sin^{-1}(x_2)) = (R_\oplus, \theta_{x_1}, \vartheta_{x_2}) \in \Theta. \quad (6)$$

The total interference at the typical SBS from the UEs in the PPP $\Theta \cap E$, with $E$ denoting the area above the horizon of the typical SBS, is defined as

$$\mathring{I} \triangleq \sum_{x \in \Theta \cap E} \frac{H_{\mathcal{MLN}_x} G(\varphi_x)}{(d_x/d_0)^\alpha}, \tag{7}$$

where $d_0$ is a normalizing constant. The simulated values are based on the spherical model, the angle $\varphi_x$ and distance $d_x$ calculated precisely for each $x \in \Theta \cap E$, and accurate i.i.d. Gaussian mixture shadowing $H_{\mathcal{MLN}_x}$.

For a more detailed analysis and planar model comparison to the spherical model, please refer to [**?**].

## III. Analysis

Given i.i.d. shadowing variables $\{H_x\}_{x \in \Phi}$, we define the process of the received signal powers at the typical location, referred to as the gain process (GP), by

$$\mathcal{G} \triangleq \{H_x G(\|x\|) : x \in \Phi\}, \tag{8}$$

where $\|x\|$ is the Euclidean distance from $o$.

The GP is a *projection process* mapping the points from $\mathbb{R}^2$ into $(0, \infty)$ and, as such, forms a nonhomogeneous PPP [**?**, Section 4.2.5].

Since the variables $\{H_x\}_{x \in \Phi}$ are i.i.d., we can denote the typical shadowing variable simply as $H$ without the subscript.

**Proposition 1** (Density of the GP)**.** *Let $F_H(\cdot)$ be the (possibly degenerate) CCDF of a fading variable $H$. The density function of $\mathcal{G}$ is given by*

$$\lambda_{\mathcal{G}}(t) = \tilde{\kappa} F_H(t)/t, \quad t \in (0, \infty), \tag{9}$$

*where $\tilde{\kappa} = \kappa/\log(2)$ and*

$$\kappa \triangleq \pi\lambda \left( \frac{\varphi_{RX} h}{\sin^2(\epsilon)} \right)^2 \tag{10}$$

*is approximately the average number of UEs inside a $-3$ dB footprint.*

*Proof.* Let $f_H(\cdot)$ be the PDF of $H$. Denote $G^{-1}(\cdot)$ as the generalized inverse of $G$, defined as $G^{-1}(y) = \inf\{x : G(x) < y\}$. According to [**?**, Eq. 4.55],

$$\begin{aligned}
&\int_t^\infty \lambda_{\mathcal{G}}(y)\, dy = \pi\lambda \mathbb{E}\left[ \left( G^{-1}(t/H) \right)^2 \right] \\
&= \pi\lambda \int_t^\infty \left( -\frac{\varphi_{\text{RX}} \sqrt{-\log(t/h)}}{D_{h,\epsilon}\sqrt{\log(2)}} \right)^2 f_H(h)\, dh \\
&= -\tilde{\kappa} \int_t^\infty \log(t/h) f_H(h)\, dh \\
&\overset{(a)}{=} -\tilde{\kappa} \left[ \log(t/h) F_H(h)|_t^\infty + \int_t^\infty \frac{F_H(h)}{h}\, dh \right].
\end{aligned}$$

In (a), we use integration by parts. The result follows by differentiating with respect to $t$ and applying the negative sign. Note that a necessary condition for this procedure is that $\int_t^\infty \log(t/h) f_H(h)\, dh$ converges for all $t > 0$.

Please refer to [**?**, Lemma 1] for throughout explanation of the geometric interpretation of $\kappa$. □

The total interference, or total received power, is defined as the sum of the GP at the footprint location $o$ of the typical SBS:

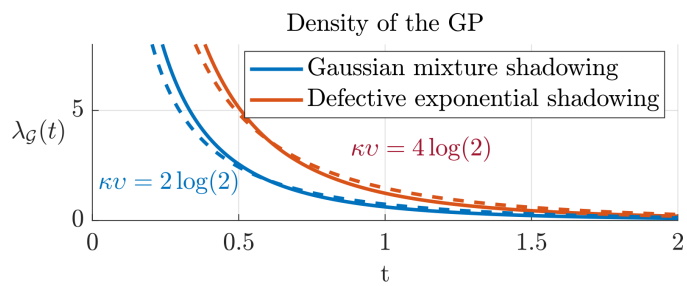
$$I \triangleq \sum_{x \in \Phi} H_x G(\|x\|) = \sum_{x \in \mathcal{G}} x. \tag{11}$$

The mean and the variance of $I$ are respectively given by

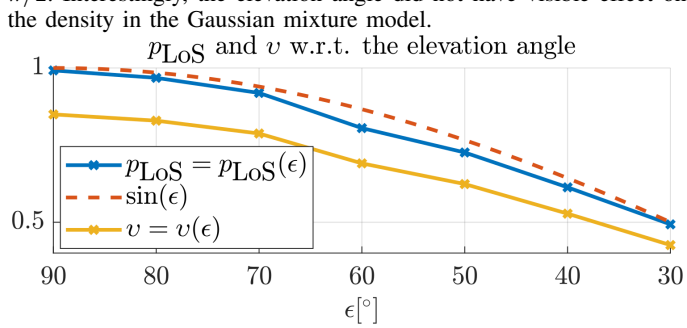
$$\mathbb{E}(I) = \int_0^\infty t\lambda_{\mathcal{G}}(t) dt = \tilde{\kappa} \int_0^\infty F_H(t) dt = \tilde{\kappa}\mathbb{E}(H), \tag{12}$$

$$\begin{aligned}
\text{Var}(I) &= \int_0^\infty t^2 \lambda_{\mathcal{G}}(t) dt = \tilde{\kappa} \int_0^\infty t F_H(t) dt \\
&= \tilde{\kappa} \frac{\text{Var}(H) + \mathbb{E}(H)^2}{2} = \tilde{\kappa}\mathbb{E}[H^2]/2.
\end{aligned} \tag{13}$$

Note that matching the first two moments of the fading distributions (2) and (3) is equivalent to matching the mean and the variance of the total interference.




(a) The density of the GP for $\kappa\upsilon \in \{2\log(2), 4\log(2)\}$ using $\epsilon = \pi/2$. Interestingly, the elevation angle did not have visible effect on the density in the Gaussian mixture model.




(b) The dependence of the shadowing parameters $p_{\text{LoS}}$ and $\upsilon$ on the elevation angle. The parameters are approximately proportional to the sine function.

Fig. 2: The density of the GP and shadowing parameters in the Gaussian mixture and defective exponential shadowing models.

### A. Laplace transform of the total received power

With the mixture exponential shadowing $H_{\text{exp}}$, for $\text{Re}(s) > 1$,

$$\begin{aligned}
\mathcal{L}_I(s) &\triangleq \mathbb{E}\left(e^{-sI}\right) = \exp\left\{ -\int_0^\infty (1 - e^{-sr}) \lambda_{\mathcal{G}}(r) dr \right\} \\
&= \exp\left\{ -\tilde{\kappa} \int_0^\infty (1 - e^{-sr}) F_{H_{\text{exp}}}(r)/r dr \right\} \\
&= \exp\left\{ -\tilde{\kappa}\upsilon \int_0^\infty (1 - e^{-sr}) e^{-r}/r dr \right\} = (1+s)^{-\tilde{\kappa}\upsilon},
\end{aligned} \tag{14}$$

which is the Laplace transform of the gamma distribution with the shape parameter $\tilde{\kappa}\upsilon$.

### B. Order statistics of the STIR and SIR processes

At the typical SBS, we denote the signal-to-interference ratio (SIR) process of the UEs as follows:

$$\Psi = \{\mathsf{Z} : \mathsf{Z} \in \Psi\} \triangleq \left\{ \frac{u}{I-u} : u \in \mathcal{G} \right\} = \left\{ \frac{H_x G(D_{h,\epsilon}\|x\|)}{I - H_x G(D_{h,\epsilon}\|x\|)} : x \in \Phi \right\}, \quad (15)$$

where $I$ is defined in (11). Similarly, the signal-to-total-interference ratio (STIR) process is defined as

$$\Psi' = \{\mathsf{Z}' : \mathsf{Z}' \in \Psi'\} \triangleq \left\{ \frac{u}{I} : u \in \mathcal{G} \right\}. \quad (16)$$

We can always recover the process from another:

$$\Psi = \left\{ \frac{\mathsf{Z}'}{1-\mathsf{Z}'} : \mathsf{Z}' \in \Psi' \right\}, \quad \Psi' = \left\{ \frac{\mathsf{Z}}{1+\mathsf{Z}} : \mathsf{Z} \in \Psi \right\}. \quad (17)$$

Let $\theta$ denote the SIR threshold for successful transmission. The event $\Psi \ni \mathsf{Z} > \theta$ is equivalent to $\Psi' \ni \mathsf{Z}' > \theta'$ with $\theta' \triangleq \theta/(1+\theta)$ and $\theta \triangleq \theta'/(1-\theta')$.

We denote $\mathsf{Z}'_{(1)} > \mathsf{Z}'_{(2)} > \mathsf{Z}'_{(3)} \dots$ as the order statistics of the STINR process $\Psi'$, such that $\mathsf{Z}'_{(1)}$ is the largest value in $\Psi'$. Through the monotonicity of the relations (17), the order statistics of the STIR process are equivalent to the order statistics of the SIR process.

**Proposition 2.** *The density of the $n$th factorial moment measure of the STIR process at the typical SBS with a narrow Gaussian antenna beam and Gaussian mixture shadow fading is (approximately) given by*

$$\mu'^{(n)}(t'_1, \dots, t'_n) = (\tilde{\kappa}\upsilon)^n \prod_{j=1}^{n} t'^{-1}_j \left(1 - \sum_{j=1}^{n} t'_j\right)^{\tilde{\kappa}\upsilon - 1}, \quad (18)$$

*whenever $t'_1 > \dots > t'_n$ and $\sum_{i=1}^{n} t'_i \leq 1$, and $0$ otherwise.*

*Proof.* The total interference can be characterized by the **gamma process** at time $\tilde{\kappa}\upsilon$ [**?**, Eq. 8] (recall (14)). Hence, the STIR process $\Psi'$ can be characterized by a **Poisson-Dirichlet distribution** $\mathrm{PD}(0, \tilde{\kappa}\upsilon)$ that has the given density [**?**, Eq. 2.3]. □

The partial densities can be derived from the density of the $n$th factorial moment measure as [**?**, Eq. 62]

$$\mu'^{(n+i)}_n(z'_1, \dots, z'_n) = \int_{z'_n}^{1} \cdots \int_{z'_n}^{1} \mu'^{(n+i)}(z'_1, \dots, z'_n, \zeta'_1, \dots, \zeta'_i) d\zeta'_1 \dots d\zeta'_i, \quad (19)$$

the support of the density being in the region $\sum_{j=1}^{n} z'_j + i z'_n \leq 1$.

The joint PDF of the $n$ strongest values of the STIR process $(\mathsf{Z}'_{(1)}, \dots, \mathsf{Z}'_{(n)})$ is given as a series expansion involving the partial densities [**?**, Eq. 64]

$$f'_{(n)}(z'_1, \dots, z'_n) = \sum_{i=0}^{i_{\max}} \frac{(-1)^i}{i!} \mu'^{(n+i)}_n(z'_1, \dots, z'_n), \quad (20)$$

for $z'_1 > z'_2 > \dots > z'_n$ and $f'_{(n)}(z'_1, \dots, z'_n) = 0$ otherwise. The upper bound for the index $i_{\max} < 1/z'_n - n$ corresponds to the non-zero terms of the series expansion.

The $n$-coverage probability that the first $n$ strongest signals reach the threshold $\theta$ is given by

$$\mathcal{P}^{(n)}(\theta) \triangleq \int_{\theta'}^{1} \cdots \int_{\theta'}^{1} f'_{(n)}(z'_1, \dots, z'_n) dz'_1 \dots dz'_n, \quad (21)$$

with $\theta' = \theta/(1+\theta)$ and $i_{\max} < 1/\theta' - n$.

The density of the $n$th factorial moment measure of the SIR process can be extracted from $\mu'^{(n)}$ [**?**, Corollary 6.1.3]:

$$\mu^{(n)}(z_1, \dots, z_n) = \prod_{j=1}^{n} \frac{1}{(1+z_j)^2} \mu'^{(n)}\left(\frac{z_1}{1+z_1}, \dots, \frac{z_n}{1+z_n}\right). \quad (22)$$

### C. SIR under interference cancellation

Let $(u_{(1)}, \dots, u_{(k)}) \subset \mathcal{G}$ represent an ordered set of points in the GP, where $u_{(1)}$ denotes the strongest signal at the typical SBS. The signals with indices in the set $[k] \triangleq (1, \dots, k), k \geq n$, are canceled from the total interference. We denote the SIR with interference cancellation as

$$\mathrm{SIR}_{n,[k]} \triangleq \frac{u_{(n)}}{I - \sum_{j \in [k]} u_{(j)}}. \quad (23)$$

Let us first study $\mathrm{SIR}_{1,[1]}$. Combining (18), (20), and (22), we can derive a closed-form for the SIR PDF of the strongest signal in the *simple coverage region* $z \geq 1$: $f_{(1)}(z) = \tilde{\kappa}\upsilon\,(z+1)^{-\tilde{\kappa}\upsilon}/z$ [1]. The second moment of the SIR is bounded by

$$\mathbb{E}(\mathrm{SIR}^2_{1,[1]}) \geq \int_1^\infty f_{(1)}(z) z^2 dz = \frac{2^{1-\tilde{\kappa}\upsilon}(\tilde{\kappa}\upsilon)^2}{(\tilde{\kappa}\upsilon - 1)(\tilde{\kappa}\upsilon - 2)}, \quad (24)$$

which is divergent for $\tilde{\kappa}\upsilon \leq 2$, *i.e.*, for less than $2\log(2)$ effective UEs inside a $-3$ dB footprint on average, the first and second moments—hence, also the variance—are infinite (or undefined). Despite the strong average SIR, the infinite variance for $\tilde{\kappa}\upsilon \leq 2$ is not desirable if we want a consistent user experience in the link quality. We demonstrate that SIC can improve user fairness.

Under perfect interference cancellation, we have the following identity in terms of the STIR process [**?**, Eq. 69]: [2]

$$\mathbb{P}(\mathrm{SIR}_{n,[k]} > \theta) = \mathbb{P}\left(\mathsf{Z}'_{(n)} + \theta' \sum_{j \in [k] \setminus \{n\}} \mathsf{Z}'_{(j)} > \theta'\right). \quad (25)$$

Following the Poisson-Dirichlet order statistics of $\mathsf{Z}'_{(1)} > \mathsf{Z}'_{(2)} > \dots$, if $\mathsf{Z}'_{(1)}$ has a finite variance, each $\{\mathsf{Z}'_{(j)}\}_{j \in [k]}$ also has a finite variance, hence $\mathrm{SIR}_{n,[k]}$ has a finite variance.

Finally, we consider the SIR under the (perfect) successive signal cancellation (SIC-SIR). A necessary condition for the successful reception of the $n$th strongest UE at the typical SBS is that the preceding $n$ signals are successively decoded and removed from the interference. Formally, $\{\mathsf{Z}'_m +$

[1] The SIR has a heavy-tailed distribution, cf. [**?**, Eq. 32].

[2] Imperfect interference cancellation is straightforward to analyze by adding an additional term in $(0, 1)$ [**?**, Eq. 6.21] to (25).

$\tau' \sum_{j=1}^{m-1} \mathsf{Z}'_j > \tau'\}$ for all $m \in \{1 \dots n\}$, where the signal detection threshold for a feasible interference cancellation is denoted as $\tau = \tau'/(1-\tau') \leq \theta$. When the first $n$ signals are successfully removed from the interference, the $n$th UE is considered covered if $\mathrm{SIR}_{n,[n]} > \theta$. If not, the SIC continues until the $\mathrm{SIR}_{n,[k]} > \theta$ or the maximum number of interference cancellation stages $K$ is reached.

**Proposition 3.** *Consider the SIC with at most $K \geq n$ interference cancellation stages. The coverage probability of the UE with $n$th strongest signal is given by*

$$\mathcal{P}_{\mathrm{SIC}}^{(n,K)}(\theta,\tau) \triangleq \sum_{k=n}^{K} \Delta_{\mathrm{SIC}}^{(n,k)}(\theta,\tau), \tag{26}$$

*where*

$$\begin{aligned}
&\Delta_{\mathrm{SIC}}^{(n,k)}(\theta,\tau) \triangleq \\
&\times \int_0^1 \cdots \int_0^1 f_{(k)}(z'_1,\dots,z'_k) \prod_{m=1}^{k} \mathbb{1}\left(z'_m + \tau' \sum_{j=1}^{m-1} z'_j > \tau'\right) \\
&\times \left(\mathbb{1}(k>n)\mathbb{1}\left(z'_n + \theta' \sum_{j\in[k-1]\setminus\{n\}} z'_j < \theta'\right) + \mathbb{1}(k=n)\right) \\
&\times \mathbb{1}\left(z'_n + \theta' \sum_{j\in[k]\setminus\{n\}} z'_j > \theta'\right) dz'_1 \dots dz'_k
\end{aligned} \tag{27}$$

*with the upper summation limit bounded by $i_{max} < 1/\tau' - 1 = 1/\tau$.*

*Proof.* The expression follows using the joint PDF of the order statistics (20). Furthermore, the first condition (with the form $\prod \mathbb{1}(\cdot)$) allows the relaxation of $i_{\max}$. Namely, a necessary condition is $z'_k + \tau' \sum_{j=1}^{k-1} z'_j > \tau'$. By simple algebra, $\sum_{j=1}^{k-1} z_j > 1 - z_k/\tau'$. Recall the condition on the non-zero terms of $\mu'^{(k+i)}$: $\sum_{j=1}^{k} z'_j + i z'_k = \sum_{j=1}^{k-1} z'_j + z'_k + i z'_k \leq 1$. The condition certainly does *not* hold if $1 - z_k/\tau' + z'_k + i z'_k > 1$. We arrive at the inequality $z'_k(-1/\tau' + 1 + i) > 0$. Divide both sides by $z'_k > 0$, and the general upper bound of $i$ follows. □

## IV. Numerical Results and Conclusions

Figures 3 and 4 depict the $n$-probabilities (21) and SIR under successive interference cancellation (SIC), *i.e.*, the SIC-SIR, for $\tilde{\kappa}\upsilon = \{2, 4\}$ (27) with $K = 3$, respectively. The signal detection threshold for a successful interference cancellation $\tau = -7$ dB. We use the values presented in Table I in the simulations ($\tilde{\kappa}$ is determined according to $\upsilon = \upsilon(\epsilon)$ to match each effective $\tilde{\kappa}\upsilon$).

The figures illustrate that the SIC-SIR can achieve comparable coverage probabilities for the strongest UE within the region $\theta \in (-7, 10)$ dB while doubling the average number of effective UEs inside a $-3$ dB footprint, denoted as $\kappa\upsilon$, and the performance of the 2nd and 3rd UEs is significantly enhanced. Consequently, a single SBS could potentially serve multiple UEs with a good performance.

Similar to (24), we can calculate an upper bound for the variance of the SIR of the strongest UE before interference cancellation for $\tilde{\kappa}\upsilon = 4$: $\mathrm{var}(\mathrm{SIR}_{1,[1]}) = \mathbb{E}(\mathrm{SIR}_{1,[1]}^2) - \mathbb{E}(\mathrm{SIR}_{1,[1]})^2 \leq 1.2$. This represents a significant improvement compared to the infinite variance for $\tilde{\kappa}\upsilon = 2$ (should we aim for a consistent reliability over the SBSs).

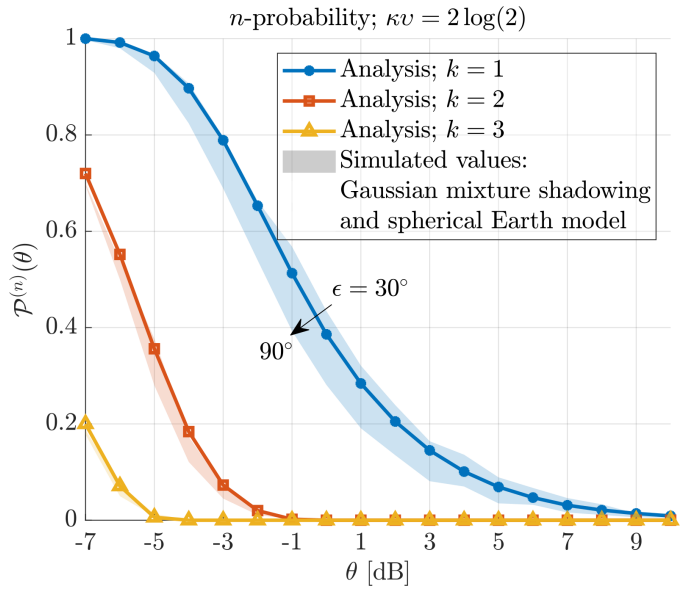



Fig. 3: The $n$-probabilities for $\kappa\upsilon = 2\log(2)$ (which is the average number of effective UEs inside a $-3$ dB footprint).

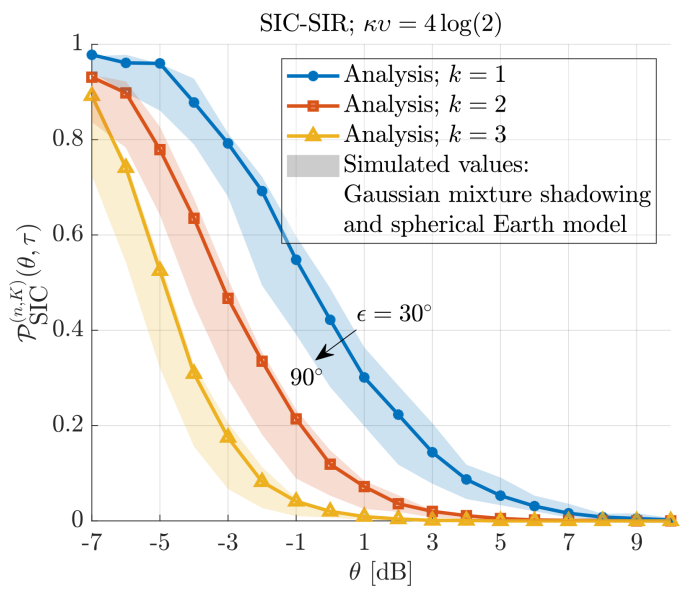



Fig. 4: The SIC-SIR for the transmitters $n \in \{1, 2, 3\}$ by doubling the average number of effective UEs inside the $-3$ dB footprints compared to Figure 3; $\kappa\upsilon = 4\log(2)$. The SIC parameter $K = 3$.

We conclude that SIC is a viable solution to reduce the considerable variability in link quality from the strongest transmitters over the satellite base stations (SBSs), while maintaining the average performance. In addition, the SIR of the second and third strongest transmitters at each SBS is significantly improved. This means a single SBS can serve multiple UEs if the SIC is implemented.

This work proposes a tractable analytical framework for the

SIC and its effect on the performance of the narrow-beam LEO network.